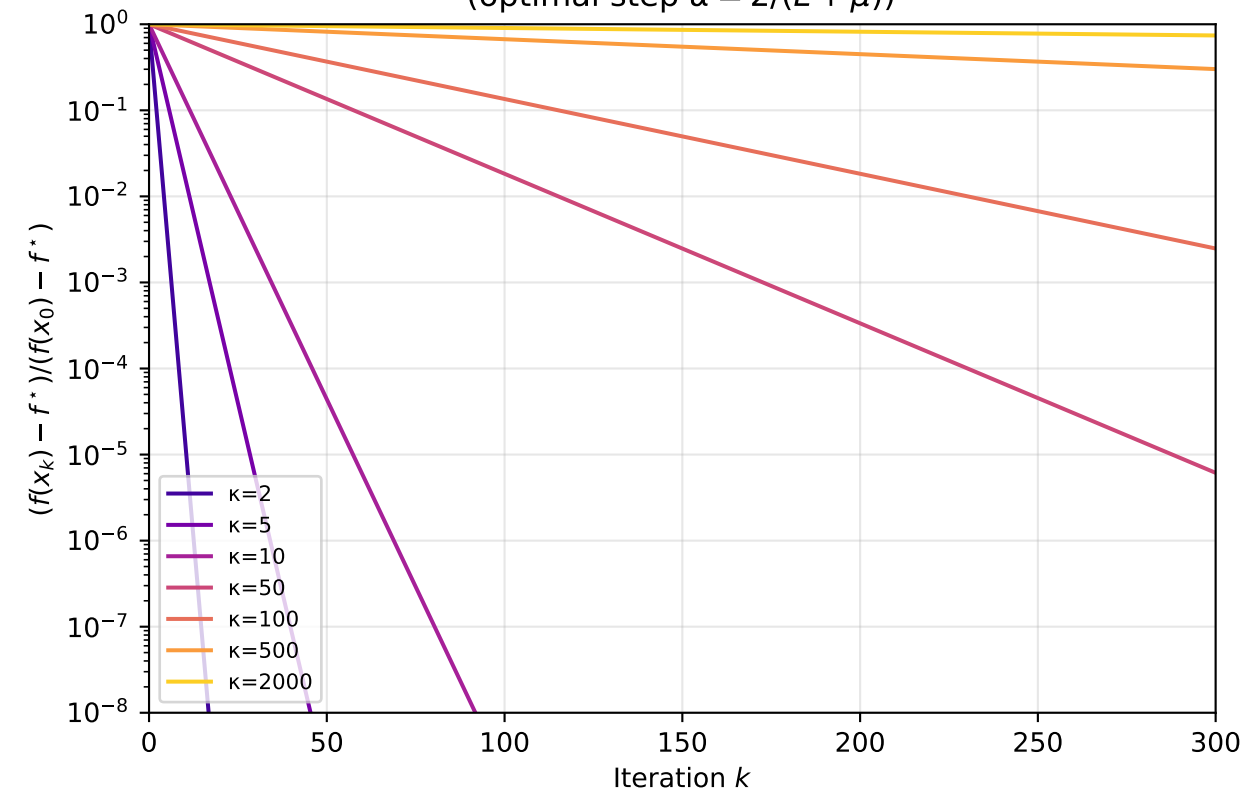
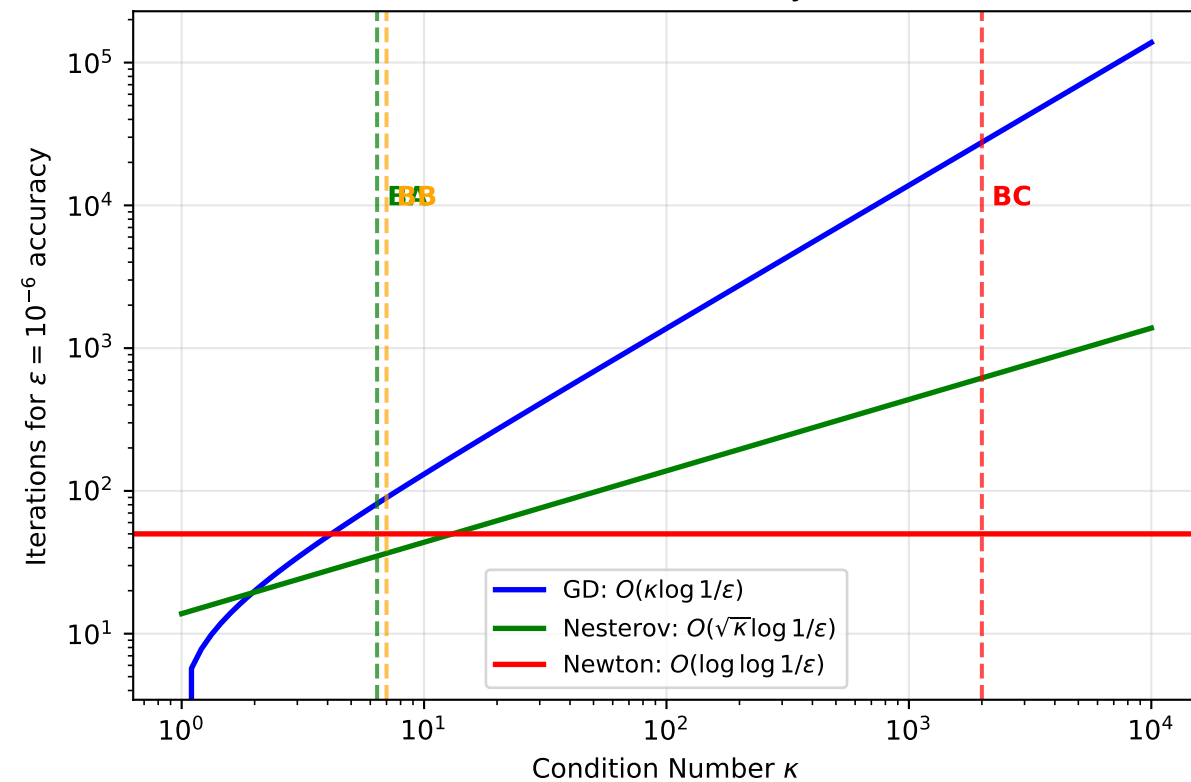


Theoretical Analysis: Condition Number as Central Determinant of Optimisation Algorithm Performance

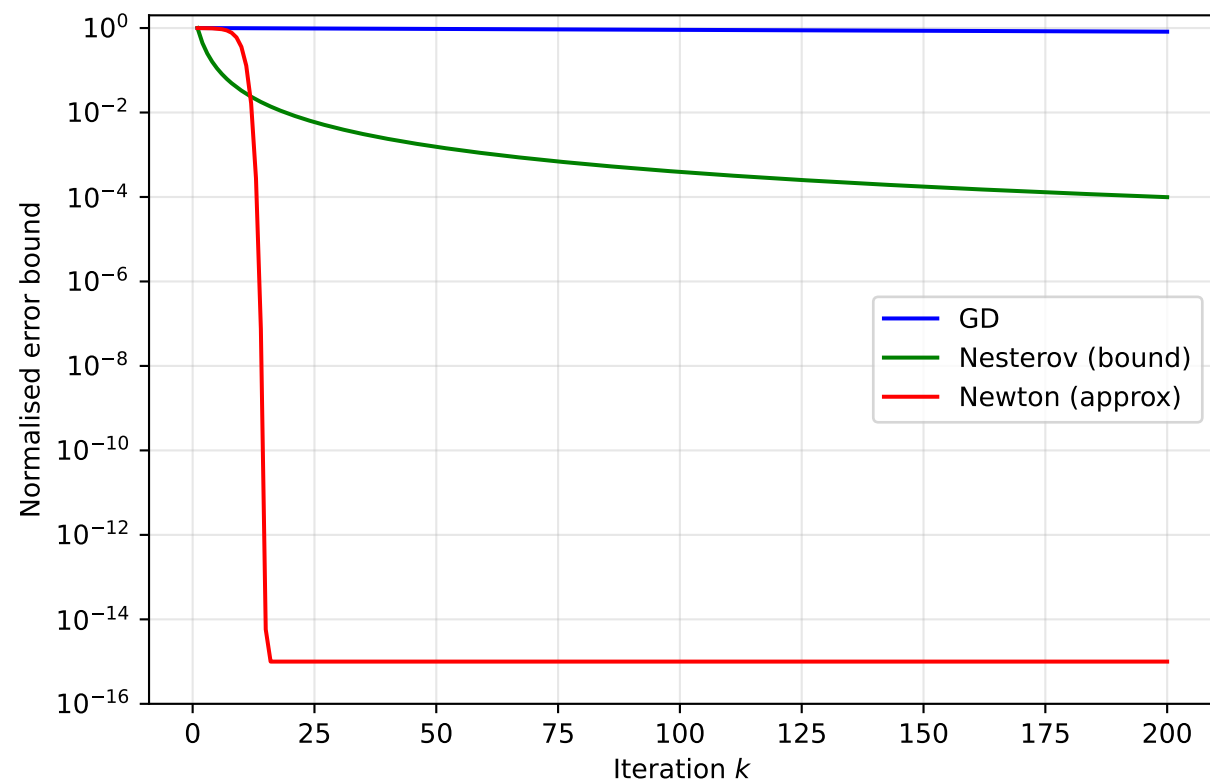
GD Linear Rate vs Condition Number (optimal step $\alpha = 2/(L + \mu)$)



Algorithm Complexity vs Condition Number ($\epsilon = 10^{-6}$ accuracy)



Convergence Rate Comparison ($\kappa = 2000$) Normalised to initial error



Final Objective on Rosenbrock (All Methods, 200 Iterations or Less)

